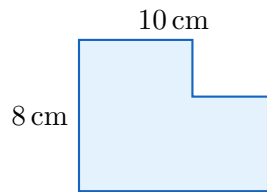


Lesson 12 - compound area by subtraction

Mild

1) Find the area by taking away the missing piece

- a) A $10\text{ cm} \times 8\text{ cm}$ rectangle with a $4\text{ cm} \times 3\text{ cm}$ corner removed.

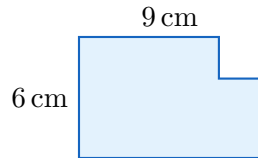


Whole rectangle = _____ cm^2

Corner removed = _____ cm^2

Area = _____ cm^2

- b) A $9\text{ cm} \times 6\text{ cm}$ rectangle with a $2\text{ cm} \times 2\text{ cm}$ corner removed.



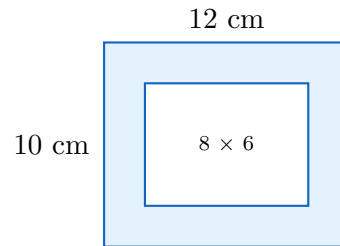
Area = _____ cm^2

Lesson 12 - compound area by subtraction

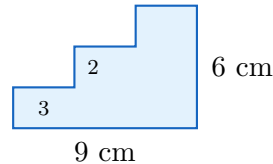
Spicy

1) Find the area of each shape by subtracting

- a) A picture frame. The outside is $12\text{ cm} \times 10\text{ cm}$. The hole in the middle is $8\text{ cm} \times 6\text{ cm}$.



- b) A 3-step staircase fits in a $9\text{ cm} \times 6\text{ cm}$ rectangle. Each step is 3 cm wide and 2 cm tall. Find its area by subtracting the steps.



2) Costing

- a) Tiles cost 15 dirhams per cm^2 . How much does it cost to tile the staircase above?

Lesson 12 - compound area by subtraction

Hot

1) Find the missing length

- a) A $14\text{ cm} \times 14\text{ cm}$ square has a notch 5 cm wide cut from one side. The shape that is left has an area of 161 cm^2 . Find the height of the notch.

2) Two methods must agree

- b) A plus-shape is 12 cm wide and 12 cm tall, with arms 4 cm thick. Find its area two ways: add three rectangles, then subtract the four corners from a $12\text{ cm} \times 12\text{ cm}$ square. Check they match.

Investigate

Omar says: "You can always split a shape into rectangles and add."

Layla says: "Sometimes taking away a piece is quicker, and for a frame it is the only sensible way."

Sketch one shape that is easier to **add**, and one that is easier to **subtract**.